# Somesh Kennedy

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## TECHNICAL SKILLS

**Languages:** Python, SQL, Bash, C++, HTML, CSS
**AI & LLMs:** LLMs, RAG, Agentic AI, LangChain, LangGraph, CrewAI, MCP, Hugging Face, Fine-Tuning, LLM Evals, Guardrails, Claude Code, Codex
**ML & Data:** PyTorch, TensorFlow, XGBoost, Pandas, PySpark, BigQuery, PostgreSQL, dbt, ETL/ELT, Data Quality, MLflow
**Backend & Cloud:** FastAPI, Flask, Django, REST APIs, Microservices, AWS (EC2, Lambda, S3, IAM, DynamoDB, SageMaker, Bedrock), Docker, Kubernetes, CI/CD, Git/GitHub, PyTest, Agile

## EXPERIENCE

**Data Engineer** — **August 2026 - Present**
*Saayam For All* — *Dallas, Texas (Remote)*

- Audited a core table's schema: documented **9** foreign-key references and cross-checked results with a peer's analysis.
- QA-tested the emergency-contact feature (**40** cases) and root-caused a **critical** silent location-fallback defect.
- Built deterministic **Python** test-data generators (**11** edge-case scenarios), validated on **PostgreSQL/PostGIS**.
- Developed **Lambda APIs** (growth, geographic, size) with **99 unit tests**; added date/country filters (merged to test).
- Integrated a **profanity-scoring API** into a **Flask/SQLAlchemy** workflow to route help requests by severity (**54 tests**).

**AI Engineer Intern** — **June 2025 - August 2025**
*WorldLink US* — *Frisco, Texas*

- Automated governance workflow triggers with **AWS Lambda**, speeding compliance processing by **45%**.
- Split agent workflows into **FastAPI microservices**, containerized with **Docker** and deployed on **AWS EC2**.
- Built **RAG**-based risk-assessment agents on **OpenAI** and **PostgreSQL**, raising audit success rate by **12%**.
- Automated prompt-evaluation benchmarks with **PyTest** and **MLflow**, shortening LLM validation cycles.
- Cut agent execution steps by **35%** with **LangGraph** pipelines using **MCP** for runtime tool discovery.

**Software Engineer Intern** — **January 2024 - March 2024**
*P9 Technologies Pvt Ltd* — *Puducherry, India*

- Engineered **Python CI/CD pipelines** to validate **data lineage** and **model quality**, cutting manual QA effort by **15%**.
- Wrote **PyTest** unit tests for core backend features, reducing code bugs by **20%**.
- Boosted retrieval precision by **25%** by constructing **hybrid vector search pipelines** on **FAISS** and **PostgreSQL**.
- Raised search response accuracy by **18%** with a **LangChain** chatbot over enterprise databases.

**Data Science Intern** — **May 2023 - July 2023**
*NIT Puducherry* — *Karaikal, India*

- Reached **94%** accuracy on predictive modeling with an ensemble of **XGBoost** and **PyTorch CNN**.
- Reduced noise in time-series signal data by **20%** with **PySpark** data-cleaning pipelines.

## PROJECTS

**Clinical Evidence Agent** | Python, SQL, Healthcare Data, LLM Evaluation | GitHub

- Built a healthcare agent on **synthetic** patient data: cited answers from **MedlinePlus** plus a guarded read-only SQL tool.
- SQL gateway blocks identifiers and writes (SQLite authorizer); emergency rules, citation checks, injection detection.
- **94.3%** recall@5 on hard retrieval queries; **34/36** agent eval runs (12/18 on the first pass); **136 tests**.

**CRM Support Agent** | Python, Hugging Face API, Salesforce REST API | GitHub

- Built an open-source **LLM agent** for CRM work, with a **Salesforce REST API** backend and an in-memory test twin.
- Enforced safety in code: closure reasons, card/SSN blocking, a 3-change cap and an **injection quarantine**.
- Built an **18-scenario eval harness**; it exposed a prompt-injection exploit in gpt-oss-20b (**0/3**), fixed to **3/3**.

**Billing Reconciler** | Python, SQL, PostgreSQL | GitHub

- Built a read-only **CLI** that recomputes invoices from raw usage and flags **mismatches**, **duplicate** and **unbilled** usage.
- Found **240/240** planted defects with no false alarms across **214K** synthetic usage events in about **2 seconds**.
- Agreed with the billing service on all **120** invoices on **PostgreSQL 16** and caught every corrupted record (**29 tests**).

**Usage Rating & Invoicing Service** | Python, FastAPI, PostgreSQL, Docker | GitHub

- Built a **REST API** that records usage and generates monthly invoices using **decimal** math and UTC billing months.
- Made usage and invoice endpoints **idempotent** via unique constraints, so retries never double-bill.
- Wrote **45 tests** (**95%** coverage) passing on **PostgreSQL 16**, plus a requirements doc with business rules.

## EDUCATION

**Master of Science in Computer Science** — August 2024 - May 2026
*University of Texas - Dallas (Richardson, TX)* — GPA: 3.97/4.0
***Courses:*** *Secure Software Development, Big Data Management, Database Designs, Machine Learning, Artificial Intelligence*

**Bachelor of Technology in Computer Science and Engineering** — December 2020 - June 2024
*National Institute of Technology Puducherry (Karaikal, India)* — GPA: 8.24/10.0